Jiaqi Shao

LLM Agents | Agent System |

GitHub Personal Webpage Email Google Scholar

Education

Hong Kong University of Science and Technology

Doctor of Philosophy (PhD) in Electronic and Computer Engineering **2023 Fall – Present**

- Supervisor: Prof. Wei Zhang (HKUST) (Mentor: Prof. Bing Luo (DKU))

The Chinese University of Hong Kong, Shenzhen

Bachelor of Engineering in Electrical and Computer Engineer **2019 – 2023**

- Stream: Computer Engineering

Experience

ByteDance | Intern (Agent Long-Run Self-Iterative Algorithm Systems)

Individual Contributor | System Design & Implementation **Jan. 2026 – Present**
Skills: **Agent System w. Harness**

- Independently led end-to-end implementation of a long-running agent and self-iterative algorithm project, from design to deployment.
- Designed and implemented a **Daemon + Rubric + Harness** architecture: daemonized scheduling and state hosting, rubric-driven evaluation/iteration, and harness-based execution orchestration and replay validation.
- Enabled three operation modes: **Auto**, **Interactive**, and **Human-Interrupt**, supporting autonomous execution, collaborative workflows, and manual takeover.
- Built robust state management across stage transitions, failure recovery, and human handoff, improving stability and controllability in long-run tasks.

Selected Research Projects

Context Folding for Agentic Multi-Turn Interactions

Lead | Optimization Algorithm Design **https://arxiv.org/pdf/2512.22733**
Skills: **Agentic RL | Context Folding | Multi-Turn Systems | Algorithm Design**

- Proposed and developed FoldAct, a algorithm for stable and efficient context folding in long-horizon RL with LLM agents, addressing critical non-stationarity in context summary policies by introducing separated loss computation, context consistency regularization, and selective segment training to overcome gradient dilution, self-conditioning collapse, and high compute cost.
- Achieved stable long-horizon policy optimization with context folding, enabling up to 5.19× training speedup for complex search agents, validated in challenging sequential reasoning and decision-making tasks.

SeekBench: Benchmarking Epistemic Competence in LLM Search Agents

Lead | Framework Design & Methodology **ICLR 2026 (Accepted), arXiv:2509.22391**
Skills: **Benchmark Design | Evaluation Methodology | Quantitative Analysis | Research Leadership**

- Developed standardized benchmark framework evaluating epistemic competence of LLM search agents. Led design of annotation schema deconstructing agent trajectories into functional types and quality attributes.
- Defined three core metrics—*Groundedness*, *Recovery*, and *Calibration*—with quantifiable frameworks (RQI, ERF, CE) for assessing reasoning-evidence integration, adaptation to low-quality information, and evidence-aware decision-making.

Beyond Right to be Forgotten: Managing Heterogeneity Side Effects Through Strategic Incentives

First Author | Incentive Mechanism Design & Theoretical Analysis **ACM MobiHoc 2025**
Skills: **Federated Unlearning | Incentive Mechanism Design | Game Theory | Theoretical Analysis**

- Studied heterogeneity side effects in federated unlearning under non-IID data, where removing a client can disproportionately degrade similarly distributed remaining clients and destabilize participation incentives.
- Developed a theoretical framework and a Stackelberg-game-based payment mechanism to retain crucial clients, improving global stability by up to 6.23%, reducing worst-case client degradation by 10.05%, and achieving up to 38.6% runtime efficiency over full retraining.

MorphAgent: Self-Evolving Multi-Agent Collaboration Platform

Co-First Author | Architecture Design & System Implementation **ICML-MAS, 2025**
Skills: **System Architecture | Multi-Agent Systems | Adaptive Algorithms**

- Designed decentralized, adaptive collaboration platform enabling LLM-based agents to dynamically evolve roles without predefined structures. Developed two-phase framework (warm-up profile optimization and continuous task execution) with core metrics (RCS, RDS, TRAS) for quantifying role evolution. Demonstrated significant improvements in task performance, domain transfer, and fault robustness across code generation, reasoning, and mathematical benchmarks.

Duke Kunshan University

Research Mentor | Research Intern **Feb. 2025 – Present**
Skills: **Research Mentorship | RAG & Memory Systems | Human-AI Collaboration**

- Lead and mentor 10+ undergraduate students in research projects focused on LLM narrative design and interactive story systems. Guide students through the complete research pipeline: project initiation, code implementation, user experiments, and paper writing. Related paper submitted to CHI 2026 and advanced to Round 2 review.
  1. **Research Direction 1:** LLM-based NPCs with long-term memory—investigate character behavior consistency and adaptability in multi-turn interactions. Design semantic retrieval (Retrieval-Augmented Generation, RAG) and memory systems to enable NPCs to maintain behavioral continuity and narrative consistency across multi-turn dialogues.
  2. **Research Direction 2:** LLM-based narrative system design—develop a hybrid framework integrating human-authored structures with generative AI, constructing a runtime-adaptive narrative engine that dynamically adjusts narrative behavior through semantic alignment mechanisms among player input, author intent, and model generation. Implement a feedback-driven long-term evolving optimization framework enabling the system to self-evolve across multi-turn interactions and cross-scenario contexts, improving story coherence.

MASArena: Benchmarking Framework for Multi-Agent Systems

Principal Lead | Architecture and Framework Design **May 2025 – July 2025**
Skills: **Framework Architecture | Experiment Orchestration | Open-Source Development**

- MASArena is an open-source, modular benchmarking framework for standardized experiment, comparison, and visual analytics in both single-agent and multi-agent systems, co-developed with Westlake University LINs-Lab.
- **Key Responsibilities and Achievements:**

1. Led and developed the overall framework design and implementation, developing a scalable modular architecture supporting plug-and-play integration of agents, tools, and datasets.
2. Developed visualization and real-time monitoring modules, comprehensive evaluation and reproducibility pipeline.
3. My detailed code contributions: https://github.com/LINs-lab/MASArena/graphs/contributors

Other Projects

FedKit: Enabling Cross-Platform Federated Learning for Android and iOS

- Developed FEDKIT, which pipelines Cross-Platform FL for Android and iOS development by enabling model conversion, hardware-accelerated training, and cross-platform model aggregation.
- Implemented workflow supporting flexible federated learning operations (FLOps) in production, facilitating continuous model delivery and training.
- Collaborated with Prof. Luo, DKU undergraduate students Sichang He (lead), Beilong Tang, and Boyan Zhang, as well as collaborators Xiaomin Ouyang (UCLA) and Daniel Nata (Flower).
- Work accepted at IEEE INFOCOM 2024 Demo.

FedCampus: Privacy-preserving Mobile Application for Smart Campus

- Designed and implemented a real-world mobile application leveraging federated learning and differential privacy for smart campus services.
- Deployed 100 customized smart watches for participants at DKU and developed FedCampus APP for Android and iOS.
- Integrated privacy-preserving analytics to enable data-driven campus improvements without compromising individual user data.
- Video demonstration available online showcasing the application's capabilities.

Edge-based Cross-device Federated Learning Prototypes

- Developed prototypes supporting Mobile and IoT devices operating at WiFi and USRP-based 4G/5G wireless networks.
- Collaborated with Prof. Luo and students from CUHKSZ to implement cross-device federated learning solutions.
- Focused on edge computing and distributed machine learning in heterogeneous network environments.

Teaching

Teaching Assistant

- ELEC3120: Computer Communication Networks (HKUST, Spring 2024)
- ELEC3300: Introduction to Embedded Systems (HKUST, Fall 2024)
- Vector Space Methods with Applications | ECE 586K (DKU, Spring 2025)